

Solid waste management

- ❑ Types of solid waste
- ❑ Sources of solid waste
- ❑ Impact of solid waste/Issues
- ❑ Prevention of solid waste pollution

Types of solid waste

- Some of the major types of solid waste are as follows:
- a. Municipal Solid Waste (MSW)
- Hazardous Wastes,
- Industrial Wastes,
- Agricultural Wastes,
- Bio-medical Wastes
- Organic/inorganic in nature
- Biodegradable/ non biodegradable
- E-waste
- Nuclear waste
- Waste from natural disasters/wars and conflicts

Sources

- ❑ Residential areas, institutions, restaurants , hotels generate MSW
- ❑ Factories such as textile mills, paper, chemical industries, breweries, slaughter house, power plants etc.
- ❑ Oil spills/leaks, water used for cleaning tankers
- ❑ Construction works, Mining, quarrying, dredging.
- ❑ Discarded food material
- ❑ Nuclear power plants, nuclear weapons.
- ❑ Wars and conflicts
- ❑ Ship-breaking
- ❑ Plastics and e- wastes

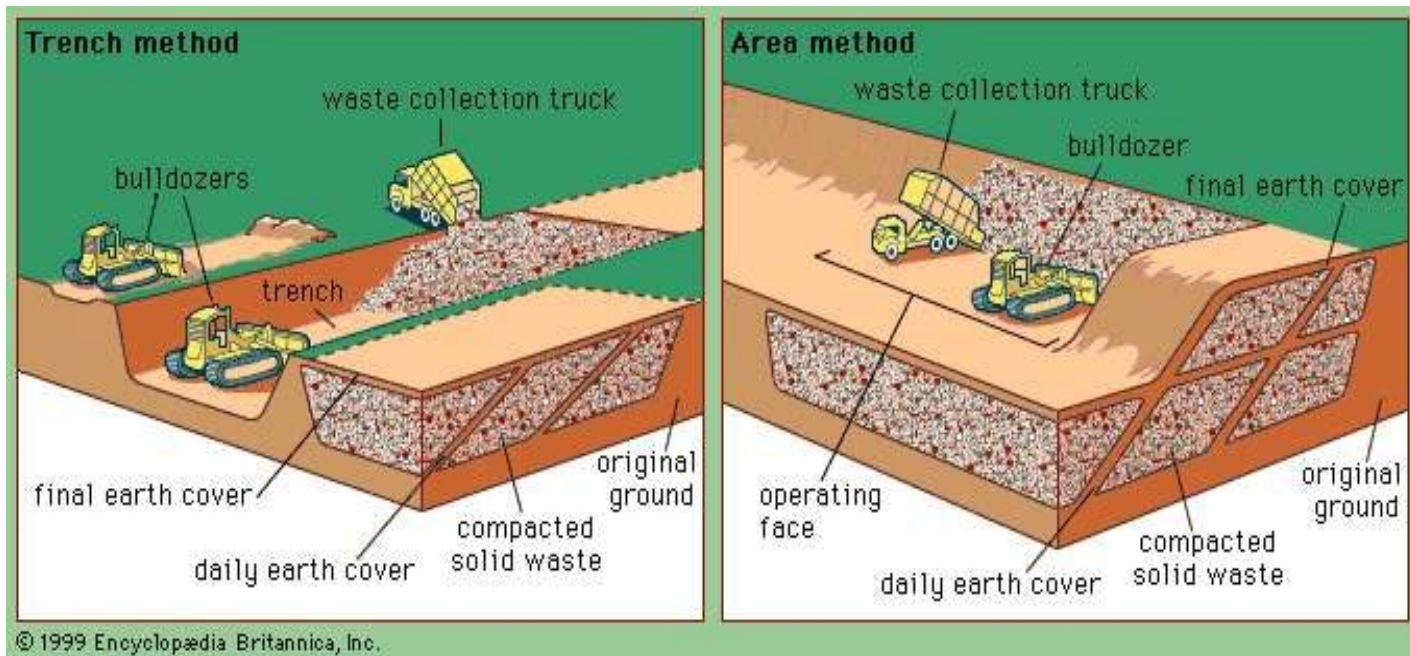
Issues

- ❑ The major problems affecting solid waste management are unscientific treatment, improper collection of waste, and ethical problems. This in turn leads to hazards like environmental degradation, water pollution, soil pollution, and air pollution.
- ❑ In most of the developing and under-developed countries sorting of solid waste right at the source is not been practiced.
- ❑ Most of the solid wastes are shifted to landfills without separation.
- ❑ And are burnt
- ❑ Ships carrying toxic materials dump the waste on an unsuspecting poor countries

Treatment of solid waste

- ❑ The collecting, treating, and disposing of solid material that is discarded because it has served its purpose or is no longer useful
- ❑ Proper solid-waste collection is important for the protection of public health, safety, and environmental quality. It is a labour-intensive activity.
- ❑ Once collected, municipal solid waste may be treated in order to reduce the total volume and weight of material that requires final disposal. This can be achieved by burning in incinerators. Incinerators that recycle heat energy in this way are called waste-to-energy plants.
- ❑ Another method of treating municipal solid waste is composting, a biological process in which the organic portion of refuse is allowed to decompose under carefully controlled conditions. Microbes metabolize the organic waste material and reduce its volume by as much as 50 percent. The stabilized product is called [compost](#) or [humus](#).
- ❑ The decomposable materials in refuse are isolated from glass, metal, and other inorganic items through sorting and separating operations.
- ❑ Shredding or pulverizing reduces the size of the waste articles, resulting in a uniform mass of material.

- Pulverized waste is ready for composting either by the open windrow method or in an enclosed mechanical facility.
- Sanitary landfill



Land fill



- One of the by-products of this decomposition is [methane](#) gas. Methane is poisonous and explosive when diluted in the air, and it can flow long distances through porous layers of soil.
- A highly contaminated liquid called [leachate](#) is another by-product, reaches the groundwater or seeps out onto the ground surface, serious [environmental pollution](#) problems can occur, including the possible contamination of drinking-water supplies.

Reuse- recycle

- Separating, recovering, and reusing components of solid waste that may still have economic value is called recycling.
- Recovered broken [glass](#) can be crushed and used in asphalt pavement. Colour-sorted glass is crushed and sold to glass manufacturers as cullet, an essential ingredient in glassmaking. [Steel](#) cans are baled and shipped to steel mills as scrap, and [aluminum](#) is baled or compacted for reuse by smelters.

Separation

- Before any material can be recycled, it must be separated from the raw waste and sorted. Separation can be accomplished at the source of the waste or at a central processing facility. Source separation, also called curbside separation.
- In lieu of source separation, recyclable materials can be separated from garbage at centralized mechanical processing plants.
- The best practice, as now recognized, is to have citizens separate refuse into a limited number of categories, including newspaper; magazines and other wastepaper; commingled metals, glass, and plastics; and garbage and other non-recyclables.

Radioactive waste

□ **Radioactive waste** is a by-product of various **nuclear** technology processes. Industries generating **radioactivewaste** include **nuclear** medicine, **nuclear** research, **nuclear** power, manufacturing, construction, coal and rare-earth mining and **nuclear** weapons reprocessing.

Q. Who is the Father of Radioactivity

1. Marie Curie
2. Antoine Henry Becquerel
3. Pierre Curie

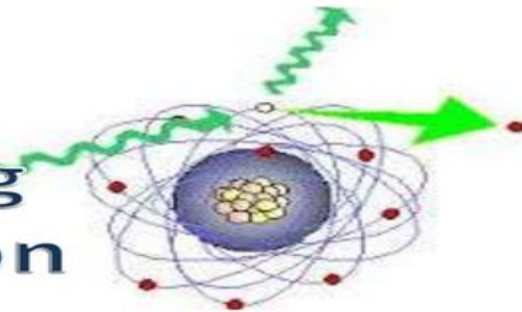


Q. Who is the Father of Indian Nuclear Programme

1. Dr. Satyendra Nath Bose
2. Dr. APJ Abdul Kalam
3. Dr. Homi Jehangir Bhabha



Types OF Ionizing Radiation



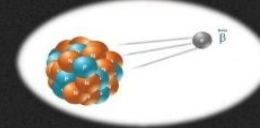
Alpha Radiation

- Emission of alpha particle
- Due to charge, it interacts strongly with matter
- Unable to penetrate
 - * sheet of paper
 - * outer layer of dead skin cells, but can if ingested in food

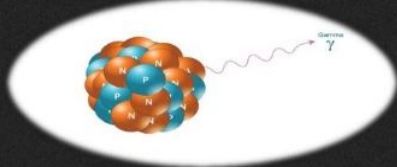


Beta Radiation

- Emission of electron or positron.
- Stopped by
 - * aluminium foil
 - * thick piece of plastic
 - * stack of papers
- Penetrates skin a few centimetres .
- Main threat: internal emission from ingested material.



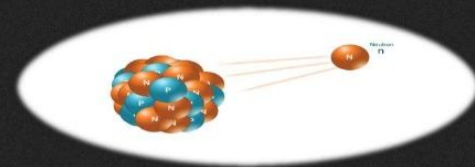
Gamma Radiation



- Emission of high energy wave.
- Can penetrate into body.
- Can be stopped by lead or depleted uranium.
- The most energetic radiation.



Neutron Radiation



- Emission of neutrons
- Blocked by hydrogen rich material like concrete or water
- Does not ionize atom directly due to lack of charge
- Only radiation that turns other materials radioactive



Radioactive Waste

- Waste containing radioactive chemical elements
- Usually, byproducts from
 - * mining
 - * industries
 - * nuclear power industries
 - * defense
 - * medicine
 - * scientific research



SOURCES

- Nuclear fuel cycle
- Nuclear weapons decommissioning
- Legacy waste
- Medical
- Industrial
- Naturally occurring radioactive material (NORM)
- Coal ,gas and oil

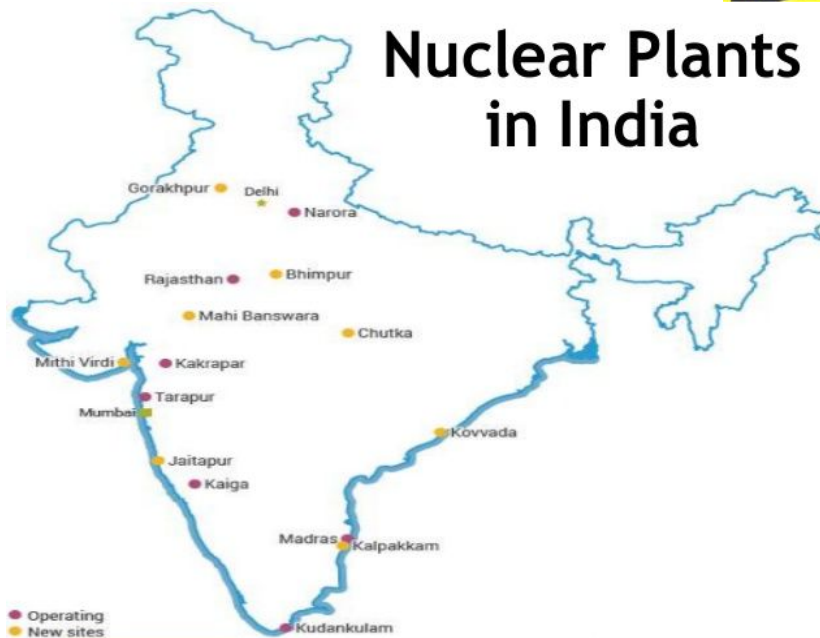


NUCLEAR FUEL CYCLE

- The fission products removed from the fuel are a concentrated form of high-level waste as are the chemicals used in the process.
- In the United States, this used fuel is usually "stored", while in other countries such as Russia, the United Kingdom, France, Japan and India, the fuel is reprocessed to remove the fission products, and the fuel can then be re-used.



Nuclear Plants in India



NUCLEAR WEAPONS DECOMMISSIONING

- Waste from nuclear weapons decommissioning is unlikely to contain much beta or gamma activity other than tritium and americium.
- It is more likely to contain alpha-emitting actinides such as Pu-239 which is a fissile material used in bombs, plus some material with much higher specific activities, such as Pu-238 or Po.



MEDICAL

- Radioactive medical waste tends to contain beta particle and gamma ray emitters
- Y-90, used for treating lymphoma (2.7 days)
- I-131, used for thyroid function tests and for treating thyroid cancer (8.0 days)
- Sr-89, used for treating bone cancer, intravenous injection (52 days)



Man-Made Sources

- Exposure is mostly through medical procedures like X-ray diagnostics.
- Radiation therapy is usually targeted only to the affected tissues.



INDUSTRIAL

- Industrial source waste can contain alpha, beta, neutron or gamma emitters. Gamma emitters are used in radiography while neutron emitting sources are used in a range of applications, such as oil well logging.



CLASSIFICATION

- Low-level Waste
- Intermediate Level Waste
- High Level Level



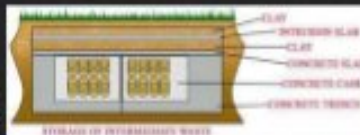
Low Level Waste

- Low level waste (LLW) is generated from hospitals and industry, as well as the nuclear fuel cycle.
- Materials that originate from any region of an Active Area are commonly designated as LLW
- LLW is suitable for shallow land burial



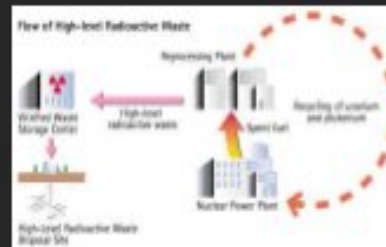
Intermediate level waste

- Contains higher amounts of radioactivity and in general require shielding, but not cooling
- Includes resins, chemical sludge and metal nuclear fuel cladding, as well as contaminated materials from reactor decommissioning.
- It may be solidified in concrete or bitumen for disposal.



High Level Waste

- HLW is produced by nuclear reactors.
- HLW worldwide is currently increasing by about 12,000 metric tons every year
- In 2010, there was very roughly estimated to be stored some 250,000 tons of nuclear HLW
- that the main proposed long-term solution is deep geological burial, either in a mine or a deep borehole.



Effects

- Accidents
- Scavenging
- Transportation
- Health Effects
- Expense
- Somatic Effects
- Genetic effects
- Naturally occurring ionizing



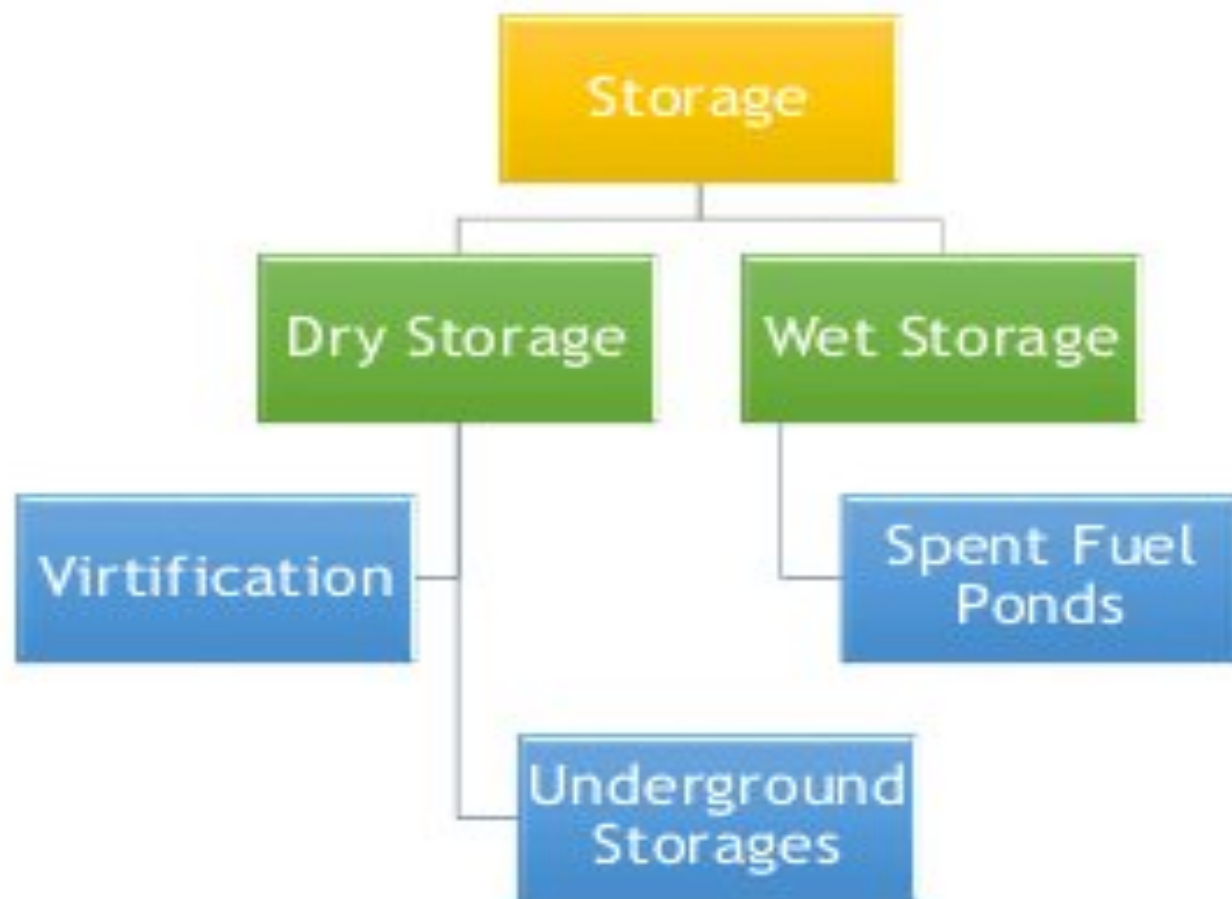
Naturally Occurring Ionizing

- Also called background radiation.
- Sources of background radiation include cosmic rays from the Sun and stars, naturally occurring radioactive materials in rocks and soil, radionuclide normally incorporated into our body's tissues, and radon and its products, which we inhale.



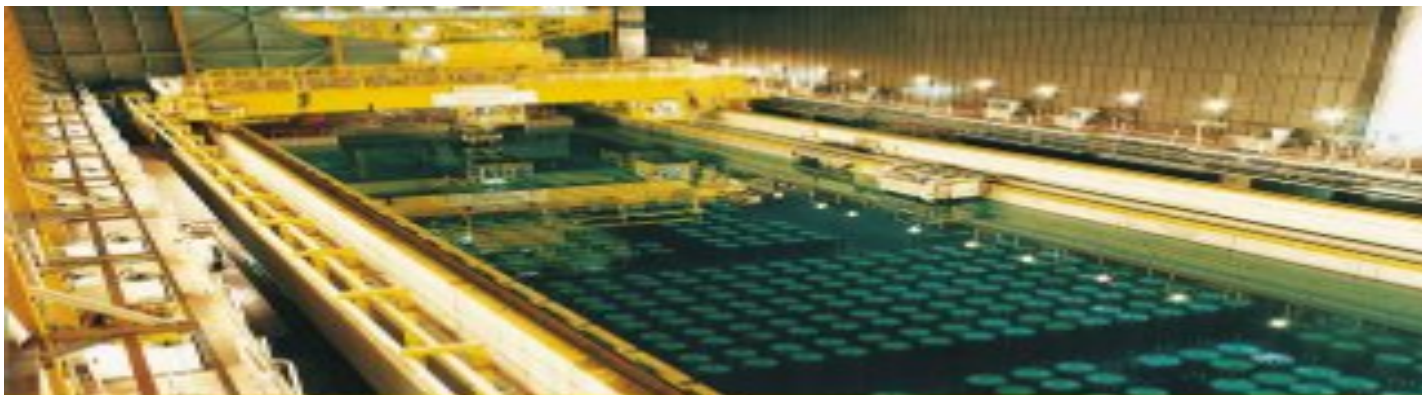
Storage of Spent Fuel

- Two different strategies are used for spent nuclear fuel:
 - ✓ Reprocessing for new fuel
 - ✓ Storing for future needs
- France, China, India, Japan, Russia reprocesses the fuel.
- USA, Finland, Sweden, Canada have opted for direct disposal.



Vitrification

It is a technique for the transformation of a substance into a glass (non-crystalline amorphous solid).



SPENT FUEL POOLS

- Spent fuel pools (SFP) are storage pools for spent fuel from nuclear reactor.
- The nuclear waste is stored at least 20 feet under the water.

- Methods of waste disposal

Methods

- Deep geological repositories
- Ocean dumping
- Seabed burial
- Sub-seabed disposal
- Subductive waste disposal method
- Transforming radioactive waste to non-radioactive stable waste
- Dispatching to the Sun
- Legal methods of waste disposal



GEOLOGICAL DUMPING

- Disposal in a facility constructed in tunnel, vaults or silos in a particular geological formation at least a few hundred meters below ground level.
- Such a facility could be designed to accept high level radioactive waste (HLW), including spent fuels if it is to be treated as waste.
- Yucca Mountain in USA is the recent of geological dumping.

Subductive waste disposal method

- It is one of the single viable disposal method which ensures NON RETURN of regulated material to biosphere

